Araz Mehrabani

Mechanical Engineer | FEA · Structural Dynamics · ANSYS

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PORTFOLIO

Technical
Project Examples
Google Drive

PROFILE & CORE COMPETENCIES

Mechanical engineer with industrial experience in **FE model development, structural dynamics and machinery engineering.** Hands-on work from geometry simplification and meshing through materials/BCs/loads to static, modal and harmonic analysis plus experimental validation.

TECHNICAL FOCUS

FE Modeling: ANSYS Workbench; geometry reconstruction/simplification, line-body/beam-element modeling, materials, contacts, boundary conditions and load cases.
Meshing: local refinement in critical regions, coarser meshes in low-stress zones, mesh-quality checks and solver-time optimization.
Structural Dynamics: static, modal and harmonic analysis; stress/deformation, natural frequencies, mode shapes, resonance and operating-range assessment.
Data & Validation: Python/MATLAB, NumPy/pandas/Matplotlib/SciPy, FFT/order analysis, time series; spring laboratory tests and result correlation.

PROFESSIONAL EXPERIENCE

Development Engineer — Sabaniroo, Tehran **01/2022–02/2023**

- **ANSYS** assessment of a wind-turbine tower-to-foundation connection: model setup, material/contact definition, boundary conditions, meshing, and stress/deformation evaluation of critical regions.

Mechanical Engineer — TehranRigzar, Tehran **03/2017–12/2021**

- Reconstructed the vibrating screen in **ANSYS** as a line-body/beam-element FE model after CATIA-import limitations; used locally adapted element sizes in critical regions to balance accuracy and computation time.
- Performed static/modal/harmonic evaluation of stresses, deformation, natural frequencies and mode shapes to define a resonance-safe operating range; calculated springs and validated them through laboratory testing.

Working Student – Mechanical Design — Ario Fidar Maham Co., Tehran **11/2012–05/2013**

- Recreated a 40x90 jaw crusher in **CATIA V5**, including individual components, assemblies and manufacturing drawings.

DYNAMICS & ANALYSIS

Master's Thesis (ongoing) – Dynamic Load Analysis — Python-based time-series pipeline with filter comparison, FFT/order diagnostics and 0P/1P/2P harmonic analysis for dynamic structural-load response assessment.

EDUCATION

M.Sc. Wind Energy Engineering, Hochschule Flensburg — current grade: 1.9 **03/2023–11/2026**
M.Sc. Mechanical Engineering, Hakim Sabzevari University — Control & Robotics **09/2014–01/2017**
B.Sc. Mechanical Engineering, Islamic Azad University — Structural Dynamics **09/2008–06/2013**

LANGUAGES

German: B2 (actively improving). English: C1 / IELTS 7.0. Persian: Native.